SOIL EROSION AND CONSERVATION · NOTE ER02

Rainfall — Forms, Storms, Indian Rainfall Patterns, Intensity–Duration–Frequency and Return Period

Course topics: SWCE3.5 L3, SWCE3.5 L6, SWCE3.5 L7, SWCE-501 T1, SWCE8.12 L3

Precipitation

- Precipitation is all measurable water received from the atmosphere — rainfall, snowfall, drizzle, hail and dew — discharged as liquid or solid onto land or water. Its study is **hydrometeorology**.
- **Rainfall:** drops > 0.50 mm in diameter; the largest stable drop is ≈ 6 mm (bigger drops break up while falling). **Drizzle:** drops < 0.50 mm, intensity usually < 1 mm/hr.
- **Snow:** ice crystals formed directly from vapour. **Hail:** ice lumps built by alternate freezing and melting in air currents. **Dew:** direct condensation at the soil surface, which slows soil-moisture depletion.
- Rain and snow supply most national water resources; in India, rainfall is the major contributor.

Rainfall type	Intensity
Light	Trace to 2.5 mm/hr
Moderate	2.5–7.5 mm/hr
Heavy	> 7.5 mm/hr

Storms

A storm is a meteorological disturbance of great magnitude, rate or intensity — rain, snow, hail, thunder, lightning or wind. Wind storms carry dust and sand; rain storms bring rain; thunderstorms bring lightning and thunder; **cloud bursts** are very intense, short rains over a small area.

System	Features
Depression	Low-pressure area with closed isobars, linked with storms and rain
Cyclone	Low barometric pressure; winds spiral inwards. A **tropical cyclone** is a violent revolving storm over the ocean — called a hurricane (Caribbean/Atlantic), typhoon (Japan/Far East) or cyclone (Indian Ocean)
Anticyclone	High-pressure area; winds spiral outward — clockwise in the Northern Hemisphere, counter-clockwise in the Southern
Tornado	Very intense circular whirl of small extent with strong winds, heavy rain or hail, thunder and lightning

Storm precipitation	Mechanism and rain
Frontal	Two air masses of different temperature and moisture meet. Cold front → heavy, short rain; warm front → widespread, low-intensity rain
Convective	Warm air rises; showery rain with rapid changes of intensity
Orographic	Moist air is lifted over a mountain barrier; rain increases with altitude

In hilly regions rainfall depends on elevation, local slope and its orientation, distance from the moisture source, topographic barriers and exposure.

Rainfall patterns of India

Season	Character
South-west monsoon (June–September)	The main rainy season, > 70% of annual rainfall. Enters through Kerala (May/June) in two branches — Arabian Sea and Bay of Bengal — both reaching Delhi by the end of June; a monsoon trough forms from the Bay of Bengal to Rajasthan. Heavy rain on the Western Ghats, Khasi-Jaintia hills, Vindhya-Satpura ranges and Himalayas; Cherrapunjee/Mawsynram receive the world's highest rainfall (> 11000 mm). Rainfall falls off sharply on the lee of the Western Ghats, gradually from east to west across the Gangetic plains, and is very low in Ladakh
Post-monsoon (October–mid-December)	As the monsoon retreats, low pressure over the Bay of Bengal brings north-east monsoon rain, mainly to Tamil Nadu; tropical cyclones cause intense rain and damage
Winter (mid-December–February)	Western disturbances bring light rain to the northern plains and moderate to heavy rain or snow to the Himalayas and J&K; some rain in Tamil Nadu from Bay of Bengal systems
Pre-monsoon (March–May)	Very little rain overall; convective thunderstorms in Kerala, West Bengal and Assam; occasional cyclones on the east coast

Rainfall region	Annual rainfall	Areas
Very high	> 2500 mm	Meghalaya, North-East, West Coast
High	1000–2500 mm	Eastern peninsula belt merging with the northern plains and east coast
Moderate	250–1000 mm	Punjab plains → Vindhyas → western Deccan → Karnataka
Low	< 250 mm	Western Rajasthan, Kutch, Ladakh plateau

Correction: the notes list "very high rainfall: up to 2500 mm", which overlaps the "high" class (1000–2500 mm). Very high means more than 2500 mm.

- Mean annual rainfall of India ≈ 1200 mm.
- **Coefficient of variability** = standard deviation of yearly rainfall ÷ normal (mean) rainfall. For India as a whole it is ≈ 10%, but it is high in low-rainfall regions — > 60% in west Rajasthan — bringing rainfall failure, drought and flood risk.

Term	Meaning
Isohyet	Line joining points of equal rainfall
Hyetograph	Bar graph of rainfall intensity against time
Rainfall excess (net rainfall)	Rainfall available for runoff
Mass curve (mass diagram)	Cumulative rainfall or runoff against time
Rainfall distribution coefficient	Maximum point rainfall ÷ mean basin rainfall

Rainfall characteristics that drive erosion and runoff

- **Amount:** depth collected on a horizontal surface with no infiltration, runoff or evaporation, in mm — per event, day, week, month or year.
- **Duration:** time over which rain falls, in minutes or hours.
- **Intensity:** depth per unit time (mm/hr), read from the slope of a recording rain-gauge chart. Higher intensity means more runoff and soil damage; a short storm may produce no runoff; runoff peaks when the whole area contributes.
- **Frequency:** the average interval between rains of a given intensity, or of equal or greater intensity. **Rainfall index:** the average intensity likely to be equalled or exceeded in a set period (e.g. 1 or 2 days) at a place over many years.

Intensity–duration–frequency (IDF) analysis

IDF relations are needed to design soil-conservation and runoff-disposal structures. High-intensity storms are short; low-intensity storms are long. A uniform high-intensity storm gives a higher, quicker and shorter runoff peak; a low-intensity storm gives a lower, later and longer one.

- Read recording-gauge charts in either of two ways: pick out periods of uniform intensity and their durations, or choose practical durations (5 minutes to 6 hours) and compute the intensity for each, splitting long storms into shorter durations.
- The intensity for a duration equal to the **time of concentration** controls the peak runoff of small and medium watersheds.
- **Storm patterns:** an **advanced** pattern (intense rain first) gives a lower runoff peak because infiltration is high early on; a **delayed** pattern (intense rain later) gives a higher peak because infiltration has fallen and storage is full.

$$\text{Mean intensity: } I = P/T \ (P = \text{rain in T hours})$$
$$\text{Quasi-rational relation: } i/I = (T + C)/(t + C)$$
$$\text{With } C = 1 \ (T, t \text{ in hours}): \ i = (P/T)\cdot(T + 1)/(t + 1)$$
$$\text{Indian IDF relation: } I = K\cdot T^{a}/(t + b)^{n}$$

In the IDF relation I = intensity (cm/hr), T = recurrence interval (years), t = duration (hr) and K, a, b, n are constants fitted from 40 recording rain-gauge stations in India. Actual intensity varies within a storm, and runoff estimates need the maximum short-period intensity.

Correction: the notes write "taking C = I", giving i = (P/T)(T + I)/(t + I). The constant is the number one: i = (P/T)(T + 1)/(t + 1), as in IRC:SP:13 for small catchments.

Rainfall is a **stochastic** variable — time-dependent, probabilistic and highly variable in space and time. It can be analysed statistically but not predicted exactly.

Recurrence interval (return period)

- Rainfall and runoff vary from day to day, month to month and year to year, so design uses a rainfall or runoff with a chosen probability of exceedance.
- The **recurrence interval** (return period, frequency) is the average time within which an event of given magnitude is equalled or exceeded at least once. Example: a 5-year, 30-min intensity of 40 mm/hr may be equalled or exceeded once in 5 years on average — a probability, not a certainty.

1. Collect at least 10 years — preferably 30 years — of data.
2. Take the annual maximum for each year and arrange them in descending order.
3. Rank them: m = 1 for the highest.
4. Probability of exceedance p = m/(n + 1), where n = number of years or events.
5. Recurrence interval RI = 1/p years.

- The top two ranked values are often discarded to raise confidence in the RI.
- **Design choice:** small soil and water conservation works — RI ≈ 5–10 years (needs about 30 years of data); large structures such as dams — RI ≈ 100–1000 years. Where long records are lacking, fit a probability distribution and extrapolate (extrapolation is hazardous).

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. IRC:SP:13, Guidelines for the Design of Small Bridges and Culverts (rainfall intensity relation). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Drizzle consists of drops smaller than

(a) 0.05 mm (b) 0.50 mm
(c) 2 mm (d) 6 mm

Q2. Heavy rainfall has intensity above

(a) 2.5 mm/hr (b) 5 mm/hr
(c) 7.5 mm/hr (d) 25 mm/hr

Q3. Orographic rainfall results from

(a) meeting of air masses (b) air lifted over mountains
(c) surface heating (d) cyclones

Q4. Recurrence interval by the Weibull method is

(a) n/m (b) (n + 1)/m
(c) m/(n + 1) (d) m/n

Q5. In a 30-year record, the 3rd-largest annual maximum has a return period of about

(a) 3 yr (b) 10.3 yr
(c) 15 yr (d) 31 yr

Q6. Coefficient of variability of rainfall in west Rajasthan exceeds

(a) 10% (b) 25%
(c) 40% (d) 60%

Q7. A storm with high intensity late in its duration gives

(a) lower runoff peak (b) higher runoff peak
(c) no runoff (d) the same peak

Q8. Soil-conservation works are usually designed for a return period of

(a) 1–2 yr
(b) 5–10 yr
(c) 100 yr
(d) 1000 yr

Answers

Q1 — (b) 0.50 mm
Q2 — (c) 7.5 mm/hr
Q3 — (b) air lifted over mountains
Q4 — (b) (n + 1)/m — *p = m/(n + 1); RI = 1/p.*
Q5 — (b) 10.3 yr — *(30 + 1)/3.*
Q6 — (d) 60%
Q7 — (b) higher runoff peak — *Delayed pattern.*
Q8 — (b) 5–10 yr